Aakaar 18th Edition

ConquerIT

Light Weight Concrete Problem Statement

About Us

Aakaar IIT Bombay is the annual technical festival of the Department of Civil Engineering at IIT Bombay. Since its inception in 2009, Aakaar has served as a premier platform for students across the country, offering opportunities to showcase and refine their skills at the highest level. Recognized as India's largest civil engineering festival, Aakaar hosts a range of thrilling competitions, including the International Civil Engineering Symposium, Bridge-It, Conquer-It, Bidwars, Crackathon, IDP, the Civil Engineering National Exhibition, ENVIZN, and Smart Pitch, enabling participants to develop practical knowledge and skills. As part of its knowledge-sharing initiative, Aakaar conducts numerous software workshops on tools like ETABS, Autodesk Civil 3D, Staad-Pro, BIM, and Revit. It also features LogiQ, a national-level civil engineering quiz competition, along with various fun-filled events that make learning engaging and enjoyable. Entering its 18th edition, Aakaar is set to reach new milestones with an array of recent events and competitions designed to provide aspiring civil engineers with a platform to explore, innovate, and learn about the immense scope of the field.

Introduction

Lightweight concrete is effective solution for addressing important construction challenges and promoting sustainable development. Lightweight concrete has been utilized for decades due to its reduced density and ability to lessen the overall structural load. Its lightweight nature makes it suitable for applications requiring improved seismic resistance and reduced material weight.

The compression strength of lightweight concrete quantifies its ability to resist axial loads and withstand heavy pressures without failure. This property is critical for evaluating the performance of lightweight concrete under load-bearing conditions.

Flexural strength, also known as the modulus of rupture, is the maximum bending stress a material can endure before yielding. For lightweight concrete, high flexural strength is essential for ensuring durability and reliability, particularly under dynamic loads such as live load or heavy construction equipment. Measuring both the weight and strength of lightweight concrete specimens provides valuable insights into its efficiency and applicability in diverse construction scenarios.

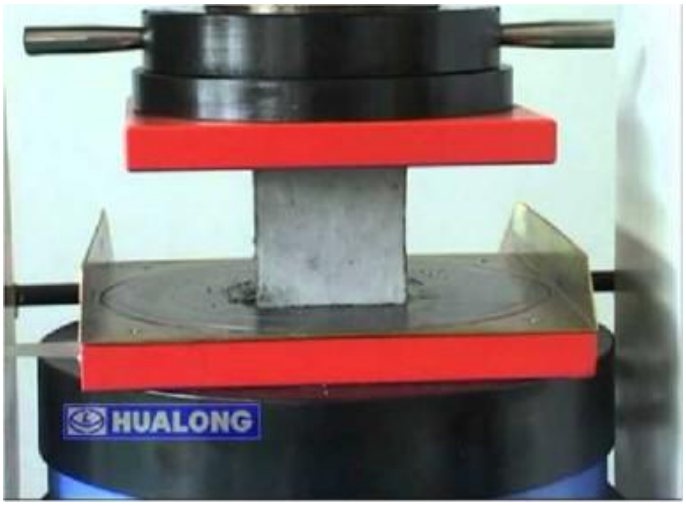
As a budding civil engineer, now is an appropriate time and AAKAAR's "Conquer IT" is the best stage to get introduced to Lightweight concrete. This year AAKAAR presents you to investigate into Lightweight concrete with high strength concrete mix.

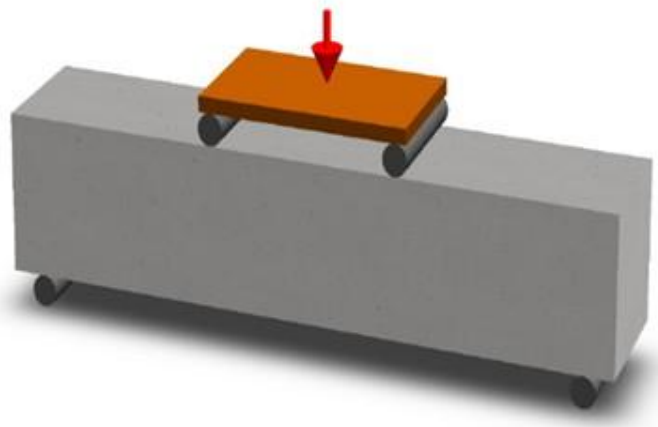
Task

1. Design and build a beam structure satisfying all the constraints mentioned below, which will be tested for Flexural Strength.
2. Cast three concrete cubes of the same mix design for the compressive strength test.

Material Constraints

For mix design, Fine aggregates can be replaced with lightweight alternatives such as crushed lightweight aggregates or manufactured sand and maximum coarse aggregate size should be less than or equal to 20mm. Participants can use any grade of concrete with any admixtures (like any air-entraining agents, fly ash, slag etc) and you can also use Glass fibres, Polypropylene Fibre in the mix design except steel fibres. Same Concrete mix should be used to cast all the specimens.





Dimensions

Strength Test

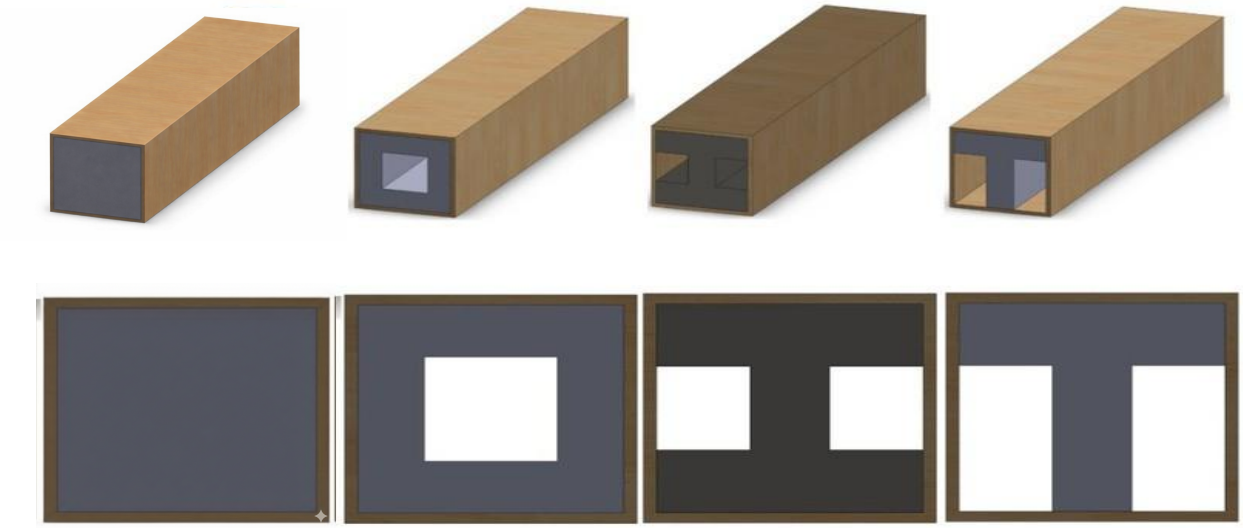
- The participants need to bring three concrete cubes made up of the same concrete mix. The cubic block will have dimensions of 150mm*150mm*150mm with an allowance of +/-5mm on all sides. Any failure to adhere to these dimensions will lead to a penalty in the total score which may further lead to disqualification of teams.

Unit Weight Test

- The concrete cubes made for compressibility test will be first used to measure unit weight, average of all the three cubes will be considered as the unit weight

Flexure Strength Test

- The beam should be in such a way that it must inscribe (i.e. the beam structure must touch 4 sides) of an imaginary cuboid of dimensions 500mm*100mm*100mm with the allowance of +/- 5mm along breadth and width and +/- 10mm along length (where 100mm is height and 500mm*100mm as the top surface)
- The structure should have at least two parallel planes, top surface should be of 100mm*500mm rectangular surface and bottom surface of 500mm length and breadth should be minimum of 30mm
- The distance between any two parallel surfaces shouldn't be less than 30mm.
- The top and bottom surfaces of the structure should be flat plane surfaces which is suitable for testing using Flexure Testing Machine (4-point loading test) although the shape of the surface may vary as per your structure.
- The structure can be of any shape satisfying the dimensions and the condition of the parallel planes mentioned above.
- The objective is to obtain a maximum flexural strength structure while minimizing the section modulus so participants are free to take any measures to reduce the section modulus (for example: Casting I shaped, T shaped, HSS beam etc..) until and unless it satisfies the constraints mentioned.



Testing Criteria:

The testing will be done in three rounds: Unit weight test and Compressive strength and flexure test.

First Round

Unit Weight Test

To measure the unit weight of concrete, weighing machine will be used and the weight of specimen will be taken upto 3 decimal places and will be used for judging.

Second Round

Compressive strength Test

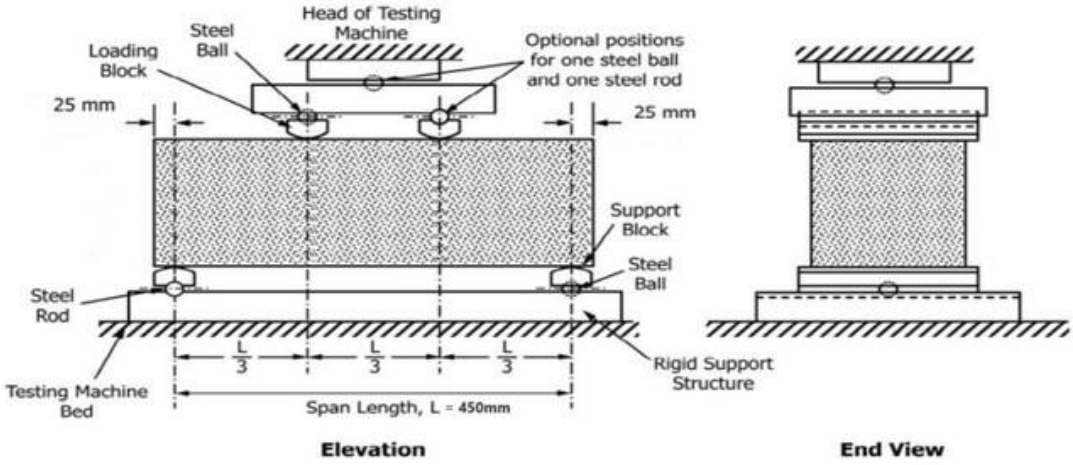
The concrete cubes will be subjected to a compressive load using a Compressive Testing Machine (CTM) at a rate of loading 30 kN/min. The compressive strength of concrete cubes will be noted to two decimal places for all three cubes. Then the average Compressive strength of all the cubes will be calculated.

Penalty: Any dimension (x) of the cube having a deviation of more than +/- 5 mm, will not be tested. Only tested cubes will be used in order to determine the average compressive strength. If all the three cubes have a deviation of more than +/- 5 mm, the team will be disqualified.

Third Round

Flexure Test

Modulus of rupture or flexural strength is calculated from the 4-point loading test where the rate of loading is 6 KN/min. The test is conducted over the length of 450mm where 25mm on each side is left to avoid unnecessary failure at supports.



Penalty: If the failure happened is not in the middle 150mm span (75mm from two sides of the centre of 500mm length) with +/- 22 mm allowance then the S3 points awarded will be zero.

Judging Criteria:

The specimen with the maximum value of total score T would be declared as the winner. In case of a tie,the preference would be given to the structure with minimum S2 value.

Scoring:

For Round 1:

S1 = 10*mean of tested cubes compressive strength (MPa)

For Round 2:

S2 = 180*(6 - w)

Where, w is the average weight of the cubes in kilograms (kg)

For Round 3:

S3 =7500*L/Z

L = Maximum Load at failure in Newtons (N)

Z = Section Modulus of the beam cross-section about the neutral axis in mm^3.

Final Score:

A score of S4 which will vary from 0 to 50 points will be allotted on the basis of your report. A detailed description of the report format is given below.

Total Score = S1+S2+S3+S4-P

Total score will be taken in order to declare a winner. Penalties(P) and report format are described below.

Report:

A report has to be submitted along with the cubes which should include the following:

- Technical mix design report which includes material specifications, mix proportions, amount and type of admixture used, casting procedure and curing procedure and mention the formulae of section modulus of your beam structure
  e.g. – Water content, Mass of cement, Mass of aggregate
- The date of casting the cubes, cylinder and beam structure
- Area moment of Inertia of the casted beam structure as to be mentioned in the report
- The report should be duly signed by a professor, who is a registered faculty member of any engineering college recognized by Government of India
- While casting, a short 2-minute video has to be shot by the participants by keeping the Beam structure and cubes side by side to ensure that participants have used the same mix to cast all the samples.
- The report should not exceed 6 sides of an A4 sheet
- Each report will be examined by the AAKAAR team and participants whose reports are not signed by the faculty member will not be allowed to take part in the competition
- Participants must explicitly calculate the Section Modulus of their beam cross-section in the report.

Penalties(P):

- If any dimension (x) of the cube having a deviation of more than +/- 5 mm, will not be tested. Only tested cubes will be used in order to determine the average compressive strength.
- If any dimension(x) of beam structure is having a deviation of morethan +/- 5 mm, then a penalty of P=10 points will be deducted for every deviated dimension.
- Any single cube exceeding 6 kg will not be tested.
- Two teams from the same college must have different report otherwise your report will not be checked i.e. S4= 0 along with a penalty of P=5 points awarded to both teams.

Aakaar 18th Edition

General Rules

- Bring from home competition so certificates will be awarded to every qualified entry.
- Students should form teams comprising of a minimum of 2 and maximum 4 members. Students from different colleges can also form a single team.
- No restriction on the use of the type of aggregates and chemical admixtures.
- No steel fibres should be used. If found, will lead to direct disqualification.
- Any curing method can be adopted.
- No paint should be added or used, it will lead to direct disqualification.
- No certificate will be provided to disqualified teams.
- The beam structure and three concrete cubes are to be brought by each team.
- While casting, a short 2 min video has to be shot by the participants by keeping the Beam structure and cubes side by side to ensure that participants have used the same mix to cast all the samples.
- This video and report should be submitted via following GOOGLE FORM before 20th February 2026
- Teams not submitting the video and report will be eliminated.
- Participants will be evaluated based on the submitted video and report, which will showcase their concrete mix design and procedure.
- Teams that do not adhere to the provided guidelines will be eliminated. Only the selected participants, who meet the criteria, will be invited to IIT Bombay with their casted specimens for final testing on the event day.
- Results of the elimination roundwill be declared by 28th February.
- Participants who qualify will be notified and invited to attend the event at IIT Bombay on the date of event.
- A registration fee of ₹300 will be applicable for the participants selected to attend the final round at IIT Bombay.
- The registration fee should be submitted via the following GOOGLE FORM by 1st March. Participants who fail to submit the registration fee by the deadline will be disqualified.
- Participants are required to bring a hard copy of the same report which they have submitted online, on the day of the event too.
- The report will be judged by AAKAAR Team.
- The AAKAAR team reserves the right to modify the competition rules in the event of unforeseen circumstances, and all decisions made by the team shall be final.
- All dates are provisional, and any changes will be communicated to participants in a timely manner

Important Dates

Events	Deadline	Forms/Link
Registration	31st January 2026	https://www.aakaariitb.org.in/compi/conquerit
Report and video Submission	20th February 2026	https://forms.gle/FsSGWJjsvgvGLfra8
Fee Submission	1st March 2026	https://forms.gle/hAf399sibRw3g9LUA
Final testing at the main event at IIT Bombay	15th March 2026	Offine Events at IIT Bombay

PRIZE WORTH : INR 25,000

For any queries,
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Mail us : competitions@aakaariitb.org.in
For more information, visit:- https://www.aakaariitb.org.in/compi/

ALL THE BEST!!
TEAM COMPETITIONS, AAKAAR 2026